A SUMMER INTERNSHIP REPORT

ON

APPLIED COMBINATORICS AND GRAPH THEORY

submitted by

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Certificate

This is to certify that ***Mr. Ankit Kumar Panda*** *has gone under a summer internship course on "****Applied Combinatorics and Graph Theory****" under my supervision at the Department of Mathematics, National Institute of Technology Calicut.*

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Abstract

This internship projects aims to understand the fundamental of applied combinatorics and graph theory. This project briefs about the various topics of graph theory in a fundamental way. It gives some insight about the utility problem and also an idea regarding the job-scheduling using bi-partite graphs.

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Chapter 1

Introduction

We start off, with a bit of history, as history intrigues us more and more.

The Seven Bridges of Konigsberg is a historically notable problem in mathematics. Its negative resolutioon by **Leonhard Euler** in 1736 laid the foundations of **graph theory** and prefigured the idea of topology.

The city of Knigsberg in Prussia (now Kaliningrad, Russia) was set on both sides of the Pregel River, and included two large islands which were connected to each other and the mainland by seven bridges. The problem was to devise a walk through the city that would cross each bridge once and only once, with the provisos that: the islands could only be reached by the bridges and every bridge once accessed must be crossed to its other end. The starting and ending points of the walk need not be the same. The difficulty was the development of a technique of analysis and of subsequent tests that established this assertion with mathematical rigor.

We will come back and discuss this problem is greater detail later on the chapter regarding *planar graphs*.

One of the most famous and stimulating problems in graph theory is the four color problem: *"Is it true that any map drawn in the plane may have its regions colored with four colors, in such a way that any two regions having a common border have different colors?"* This problem was first posed by Francis Guthrie in 1852 and its first written record is in a letter of De Morgan addressed to Hamilton the same year. Many incorrect proofs have been proposed, including those by Cayley, Kempe, and others. The study and the generalization of this problem by Tait, Heawood, Ramsey and Hadwiger led to the study of the colorings of the graphs embedded on surfaces with arbitrary genus. Tait's reformulation generated a new class of problems, the factorization problems, particularly studied by Petersen and Knig. The works of Ramsey on colorations and more specially the results obtained by Turn in 1941 was at the origin of another branch of graph theory, extremal graph theory.

Applications of Graph Theory include :-

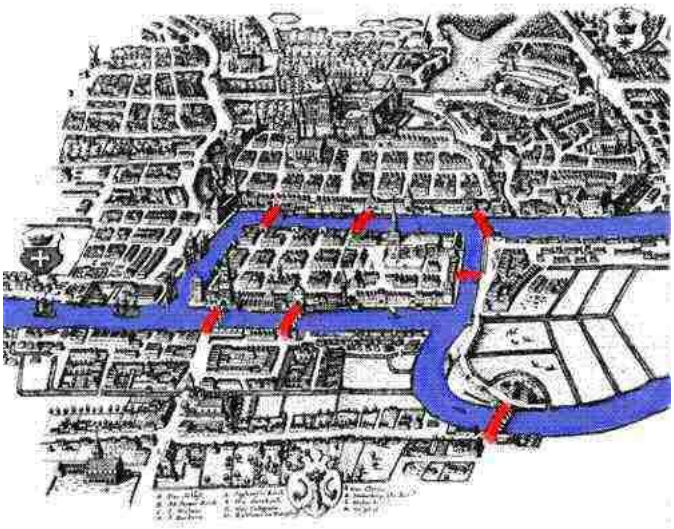


Figure 1.1: Seven Bridges of Konigsberg

1. In computer science, graphs are used to represent networks of communication, data organization, computational devices, the flow of computation, etc. For instance, the link structure of a website can be represented by a directed graph, in which the vertices represent web pages and directed edges represent links from one page to another. A similar approach can be taken to problems in travel, biology, computer chip design, and many other fields. The development of algorithms to handle graphs is therefore of major interest in computer science. The transformation of graphs is often formalized and represented by graph rewrite systems. Complementary to graph transformation systems focusing on rule-based in-memory manipulation of graphs are graph databases geared towards transaction-safe, persistent storing and querying of graph-structured data.

2. Graph-theoretic methods, in various forms, have proven particularly useful in linguistics, since natural language often lends itself well to discrete structure. Traditionally, syntax and compositional semantics follow tree-based structures, whose expressive power lies in the principle of compositionality, modeled in a hierarchical graph. More contemporary approaches such as head-driven phrase structure grammar model the syntax of natural language using typed feature structures, which are directed acyclic graphs. Within lexical semantics, especially as applied to computers, modeling word meaning is easier when a given word is understood in terms of related words; semantic networks are therefore important in computational linguistics. Still other methods in phonology (e.g. optimality theory, which uses lattice graphs) and morphology (e.g. finite-state morphology, using finite-state transducers) are common in the analysis of language as a

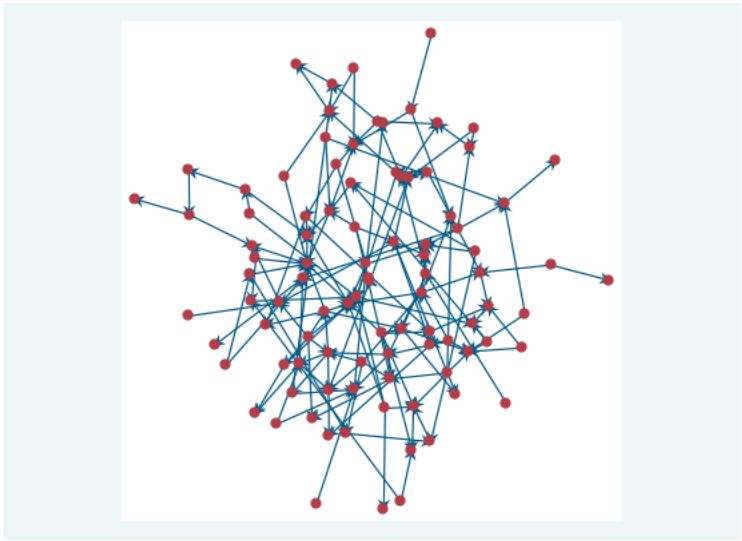


Figure 1.2: A network of computers

graph. Indeed, the usefulness of this area of mathematics to linguistics has borne organizations such as TextGraphs, as well as various 'Net' projects, such as WordNet, VerbNet, and others.

3. Graph theory is also used to study molecules in chemistry and physics. In condensed matter physics, the three-dimensional structure of complicated simulated atomic structures can be studied quantitatively by gathering statistics on graph-theoretic properties related to the topology of the atoms. In chemistry a graph makes a natural model for a molecule, where vertices represent atoms and edges bonds. This approach is especially used in computer processing of molecular structures, ranging from chemical editors to database searching. In statistical physics, graphs can represent local connections between interacting parts of a system, as well as the dynamics of a physical process on such systems. Graphs are also used to represent the micro-scale channels of porous media, in which the vertices represent the pores and the edges represent the smaller channels connecting the pores.

4. Likewise, graph theory is useful in biology and conservation efforts where a vertex can represent regions where certain species exist (or inhabit) and the edges represent migration paths, or movement between the regions. This information is important when looking at breeding patterns or tracking the spread of disease, parasites or how changes to the movement can affect other species.

Chapter 2

Algebra of Sets

2.1 Basic definations

A *set* is a well defined collection of distinct objects, like $A = \{a, b, c, d\}$. Going further we define the **product** of two sets as $A \times B = \{(a, b); a \in A, b \in B\}$. The term **cardinality** says about the number of elements or objects present in a set.(*for* $A = \{a, b, c, d\}$ *it's* 4*, and the notation used is* $|A| = 4$.). The cardinality for the product of two sets is $|A \times B| = n \cdot m$, if $|A| = n$ and $|B| = m$.

The **powerset** of any set S, written as $\wp(S)$ is the set of all subsets of S, including the empty set and S itself. An interesting property, regarding power sets tells us that, if S is a finite set, with $|S| = n$, then the number of subsets of S is $|\wp(S)| = 2^n$.

Extending this concept to accomodate the study of graphs, we discuss about the power of a set S. For example the term $A^{[k]}$ is the set of all subsets of A having k elements.That is for 2-elements subset of A we write $A^{[2]}$. With little intuition we can see that the cardinality of $A^{[2]}$ is $\binom{n}{2}$ if $|A| = n$.

Progressing, a **proper subset** of a set A is simply a set which contains some but not all of the objects in A like $B \subset A$. Likewise, an **improper subset** of a set A is subset which can be equal to the original set as in $C \subseteq A$.

2.2 Countability of Set

A map f between sets S_1 and S_2 is called a bijection if f is one-to-one and onto. In other words

If $f(a) = f(b)$ then $a = b$. This holds for all $a, b \in S_1$. For each $b \in S_2$, there is some a in S_1 such that $f(a) = b$.

Let $\mathbb{N} = \{1, 2, 3, \ldots\}$ denote the natural numbers. A set S is called countable if $S \sim T$ for some $T \subset N$.

Remark. *If S is countable and $S' \subset S$, then S is also countable, and if S is not countable then S' too is not countable.*

A example will further strengthen this concept. The integers $\mathbb{Z}$ form a countable set. A bijection from $\mathbb{Z}$ to $\mathbb{N}$ is given by $f(k) + 2k$ if $k \geq 0$, and $f(k) = 2(-k) + 1$ if $k < 0$. So, f maps to $0, 1, 2, 3 \ldots$ to $0, 2, 4, 6 \ldots$ and f maps $-1, -2, -3, -4 \ldots$ to $1, 3, 5, 7 \ldots$

2.3 Cantor Theorem

George Cantor showed that the power set of a set (whether infinite or not) always has strictly higher cardinality than the set itself. This leads to a theorem, that is **the Cantor theorem.**

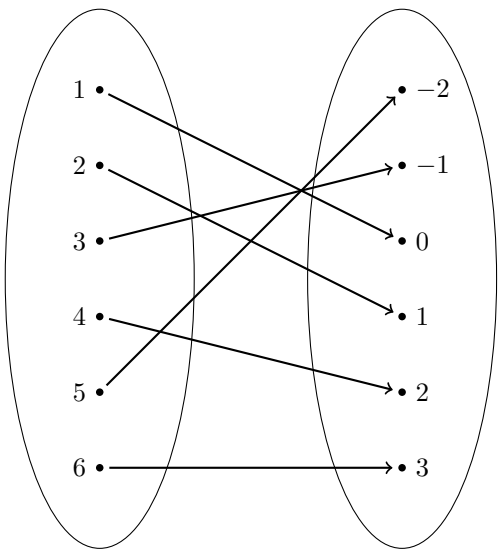


Figure 2.1: Bijection mapping for infinite sets

Theorem 1 (Cantor)**.** *For any set A, the set of all subsets of A (the power set of A) has a strictly greater cardinality than A itself. In fact, for any set S , there is no function from S* ***onto*** *its power set.*

An interesting point is to be noted here. For an *infinite set*, **bijection** might be possible. For it to start let's define and understand what countibilty of a set is.

For example, consider the set of all natural numbers $\mathbb{N} = \{1, 2, 3, \ldots\}$ and the set of all integers $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.where $\infty \notin \mathbb{N}$.

We have to show that there exists a bijection from $\mathbb{N}$ to $\mathbb{Z}$.

A way to map the elements is mapping even elements in $\mathbb{N}$ to the positive elememts in $\mathbb{Z}$, and mapping odd elements in $\mathbb{N}$ to the negative elements in $\mathbb{Z}$, as shown the figure. Maping up, we finally get,

$$\boxed{f(x) = \begin{cases} 0 & \text{if } n = 1 \\ \frac{n}{2} & \text{if } n = \text{even} \\ \frac{1-n}{2} & \text{if } n = odd \end{cases}} \tag{2.1}$$

An example as a figure 2.1 gives us a clearer picture.

Chapter 3

Basic Graph Terminology

All the notations are taken from Diestel.[1]. A **simple graph** $G = (V, E)$ is an ordered pair, where V is a non-empty set and E is a subset of the set $[V]^2$ of unirdered pairs of distinct elements of V. The set V or $V(G)$ is called the

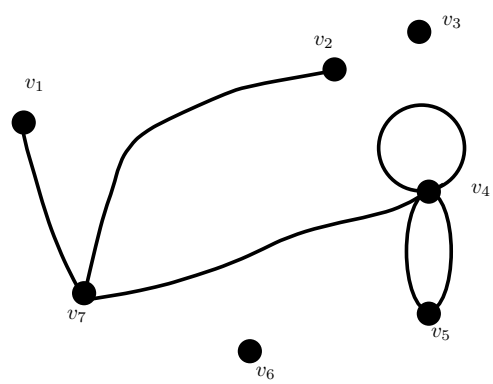


Figure 3.1: A Graph

vertex set of G and the set E or $E(G)$ is called the **edge set** of G. The vertex set in Fig 3.1 is $\{v_1, v_2, v_3, v_4, v_5, v_6\}$. The number of vertices in $V(G)$ is called **order** of the graph and is denoted by p. The **size** of the graph G is the number of edges q, in $E(G)$. Thus, a (p, q)-graph is a graph having p vertices and q edges.

An **edge** (u, w) or u, v is said to join u and v, and is usually denoted by uv. As the topic of discussion is restricted to undirected graphs, uv and vu are equal. If $uv \in E(G)$, then u and v are **adjacent**, or neighbouring vertices of G and the vertices u and v are **incident** with the edge uv. So, two edges are adjacent if they have exactly one common vertex.

A vertex $v \in G$ is **pendant**,(v_2 in the fig.3.1) if it is adjacent to exactly one

vertex of G and a vertex $v \in G$ is isolated if it is not **adjacent** to any vertex of G.(v_6 in the Fig.3.1)

A graph H is said to be a **subgraph** of a graph G, $H \subseteq G$, if all the vertices and edges of H are in G. I is the **induced subgraph** of G and for any pair of vertices $u, v \in V(G)$, if $uv \in E(G)$, then $uw \in E(I)$. Finally, a subgraph S of G is called a **spanning subgraph** if every vertex of S lies on G.That is, $S(V', E')$ is the spanning subgraph of $G(V, E)$ if $V' = V$.

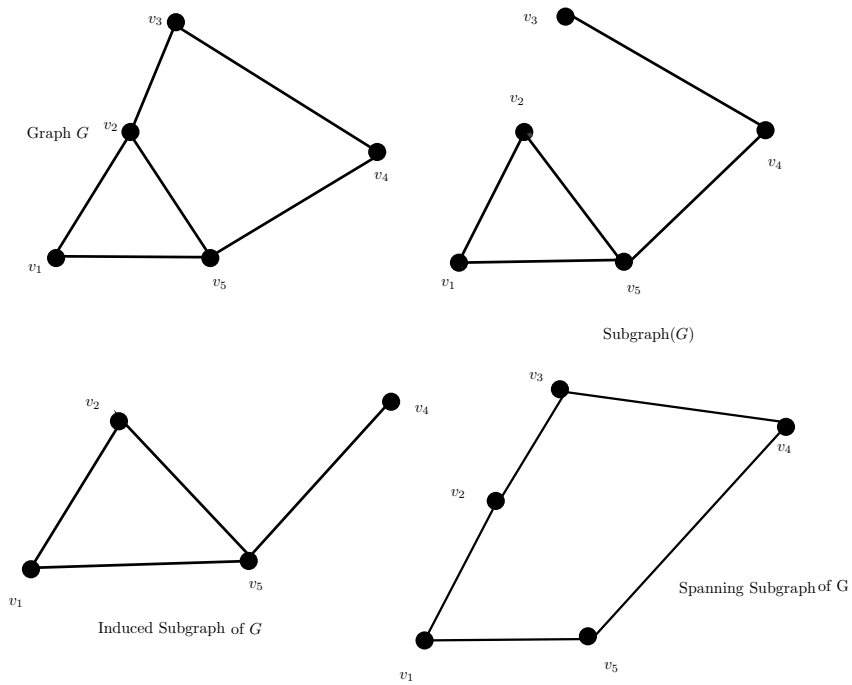


Figure 3.2: Different types of Subgraphs

A **complete graph** is a graph in which any two vertices are adjacent and is denoted by K_n, which has $\binom{n}{2}$ number of edges.

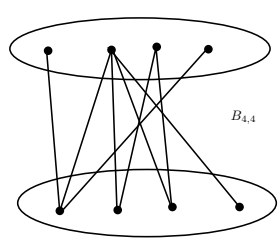


Figure 3.3: Bipartite Graph $B_{4,4}$

Line graphs $\{L(G)\}$ of a graph $G(V, E)$ having edge-set $E = \{e_1, e_2, e_3 \ldots, e_q\}$ are those which have the vertex set $V(L(G)) = \{v_1, v_2, v_3 \ldots, v_q\}$ and edge set as $E(L(G)) = \{e_i e_j : e_i$ and e_j have a common vertex $\}$.

A graph $G = (V, E)$ is **bi-partite**, if $V = V_1 \cap V_2$ and $V_1 \cap V_2 \neq \phi$ and every edge joins a vertex of V_1 to a vertex V_2 The Figure[3.3] shows an example of a bi-partite graph $B_{4,4}$. The bi-partite graph offers a solution to the *job scheduling problem.*

Chapter 4

Fundamental Theorem of Graph Theory

Let's start this chapter with a classical question, which is popularly known as '*The Handshake Problem*'.

For graph theory, we got to prove this lemma.

> In a party of people some of whom shake hands, an **even number** of people must have shaken an **odd number** of other people's hands.

To develop the intuition to prove this lemma we have to dig further and know the Fundamental Theorem of Graph Theory.

4.1 The Fundamental Theorem

4.1.1 Graph Neighborhood and Degree

The **graph neighborhood** $N(v)$ is defined as a collection of all vertices, and mathematically using set notation :-

$$N(v) = \{w \in V : wv \in E\} \tag{4.1}$$

And, the **degree of a vertex** v in a graph G is the number of vertices in the neighborhood of v.
A graph is called k**-regular** if every vertex of the graph has a degree of fixed constant k.

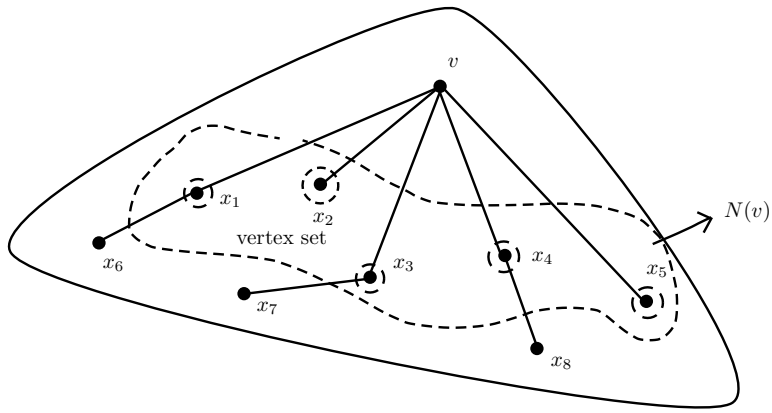


Figure 4.1: Neighborhood of v

4.1.2 Finally the Fundamental Theorem

Having established basic terms required, we can finally state and prove the fundamental theorem of graph theory.

Theorem 2 (Fundamental Theorem of Graph Theory)**.** *Let $G(v, e)$ be a graph with p vertices and q edges, then the sum of the degrees of the vertices is equal to $2q$.*

$$\sum_{i=1}^{p} d(v_i) = 2q \tag{4.2}$$

Proof. Let there are q edges in the graph $G(v, e)$.

So $E = \overbrace{\{e_1, e_2, e_3, e_4, e_5, e_6, e_7 \ldots\ldots\ldots e_q\}}^{\text{each edge contributes '2' to the degree sum } d}$

Counting the number of incident pairs (v, e) where e is an edge and vertex v is one of its endpoints, in two different ways, we get each edge contributing two times. So, each edge is contributing 2 to the degree sum, ehich implies the sum of the degrees of all the vertices is twice the number of edges. Therefore the degree sum d is $2q$. Hence the theorem is proved. □

Now let's come back to the handshake lemma. As we have understood the

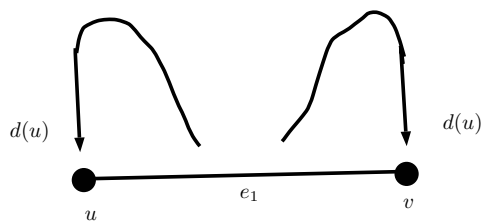


Figure 4.2: An edge gives $'2'$ to the degree

Fundamental Theorem, it might be easier to prove the lemma. To prove the lemma is basically to prove the corollary of the Fundamental Theorem, which is,

Corollary 2.1. *The number of odd degree vertices in a graph G are even in number.*

Proof. Let G be a (p, q)-graph, and consider the sum of vertices Let $V = \{v1, v2, \ldots, vp\}$ be the vertex set of G. Then,

$$\sum_{i=1}^{p} d(v_i) = 2q \tag{4.3}$$

Thus, the sum of all degrees is an even integer. Now if the vertices $\{v_1, v_2, v_3 \ldots, v_k\}$ have odd degrees and $\{v_{k+1}, \ldots, v_n\}$ have even degrees, then [from the Fundamental Theorem] $d(v_1) + \ + d(v_k) = 2m - d(v_k + 1) - \ldots - d(v_n)$ is even. Therefore, k is even. Hence there are even number of odd degree vertices.

Mathematically, this can be shown like,

$$\left(\sum_{v \in V} deg(v)_G\right) - \left(\sum_{v \in V; deg(v)_G = 2k} deg(v)_G\right) = \left(\sum_{v \in V; deg(v)_G = 2k+1} deg(v)_G\right) \tag{4.4}$$

...and as proved the result must be an even number as it's the difference of two even numbers. Since this is a sum of exclusively odd terms, there must be an even number of such terms for the sum on the right side to be even. □

Chapter 5

Paths, Cycles, Trees and Connectivity

We begin our discussion for this chapter by understanding what paths, cycles in a graph are,components of a graph,trees and then extend the topic to product of cycles, connectivity, Whitney's Theorem and conclude with a real-world example on maximum-flow minimum-cut problem.

5.1 The Road Taken

An alternating sequence $(u = v_0)e_0v_1e_1 \dots v_ie_iv_{i+1} \dots e_{n-1}(v_n = v)$ of vertices and edges, where *each edge* is incident with its preceding and succeeding vertices is a $(u - v)$ **walk**. In fig 5.1 $\{v_4, e_5, v_4, e_3, v_3, e_2, v_2, e_7, v_5, e_6, v_4\}$ is the walk.

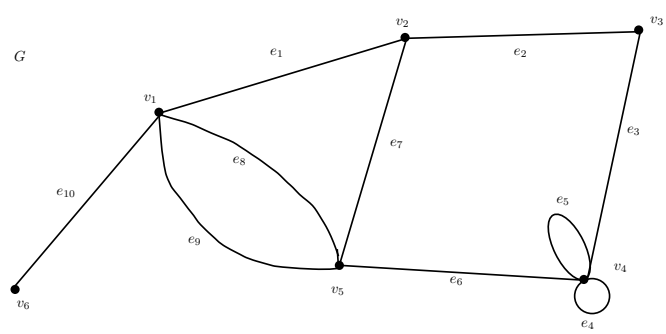


Figure 5.1: Graph with walk,path and cycle, Rouhonen.[2]

An $(u - v)$ walk is called a **trail** if *all edges* in the sequence are distinct. In fig 5.1 the $\{v_1, e_8, v_5, e_9, v_1, e_1, v_2, e_7, v_5, e_6, v_4, e_5, v_4, e_4, v_4\}$ is a trail.

A walk $(u-v)$ is a **path** if *all the vertices and edges* in the sequence are distinct. The path on n vertices is denoted by P_n. For example, $\{v_2, e_7, v_5, e_6, v_4, e_3, v_3\}$ is the path, in the fig 5.1

5.2 Cycles

A path $(u = v_0)e_0v_1e_1 \dots v_ie_iv_{i+1} \dots e_{n-1}(v_n = v)$ in which $u = v$ is called a **cycle.** (*or closed path*), and is denoted by C_n. The **distance** between two vertices u and v is $d(u, v)$ which is the number of edges in the shortest path connecting u and v.

If there is *no path between* u and v, then we define $d(u, v) = \infty$.

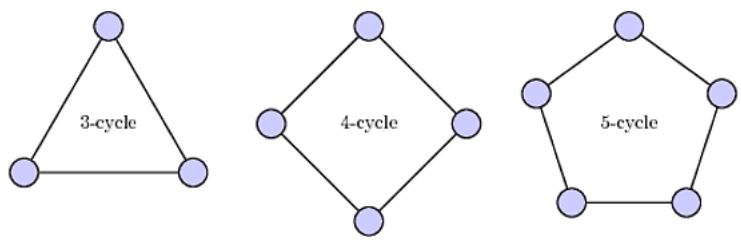


Figure 5.2: Cycle graphs

Using these concepts we define a **connected graph** G as one in which the distance between any two vertices of G is finite. A **disconnected graph** is one which there exists pair of vertices u, v for which $d(u, v) = \infty$. *A maximal connected subgraph* of a graph G is a component of the graph.

5.2.1 Graph Algebra

The **union of two graphs** $G_1 = (V(G_1), E(G_1))$ and $G_2 = (V(G_2), E(G_2))$ is the graph $G = G_1 \cup G_2$ with the vertex set $V(G) = V(G_1) \cup V(G_2)$ and the edge set $E(G) = E(G_1) \cup E(G_2)$.

Similarly, the **intersection** of graphs $G = G_1 \cap G_2$ is the graph with the vertex set $V(G) = V(G_1) \cap V(G_2)$ and the edge set $E(G) = E(G_1) \cap E(G_2)$.

Continuing this, the **sum of two graphs** $G_1 = (V(G_1), E(G_1))$ and $G_2 = (V(G_2), E(G_2))$ is denoted by $G_1 + G_2$, which is a graph with the vertex set $V(G_1) \cup V(G_2)$, and the edge set $E(G_1) \cup E(G_2) \cup \{uv : u \in G_1, v \in G_2\}$.

The **cartesian product** of two graphs , $G_1 = (V(G_1), E(G_1))$ and $G_2 = (V(G_2), E(G_2))$ with disjoint point set v_1 and v_2 and edge set e_1 and e_2 is the graph with point set v_1v_2 and $u = (u_1, u_2)$ adjecent with $v = (v_1, v_2)$ where ($u_1 = v_1$ and u_2 adjeacnt to v_2) or ($u_2 = v_2$ and u_1 adjacent to v_1).

The complement of a graph $G = (V, E)$, is the graph $\overline{G} = (V, [V]^2 - E)$. Alternatively the complement of a graph G is $\overline{G}$ with vertex set $V(\overline{G}) = V(G)$ and edge set $E(\overline{G}) = \{uv : u, v \in G$ and u is not adjacent to $v\}$

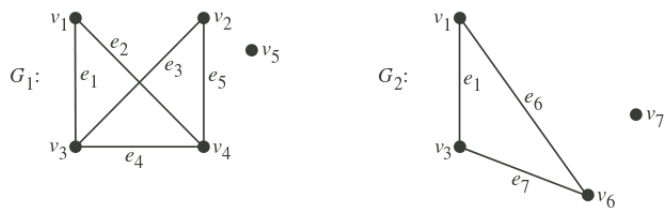


Figure 5.3: Graphs for algebraic operations

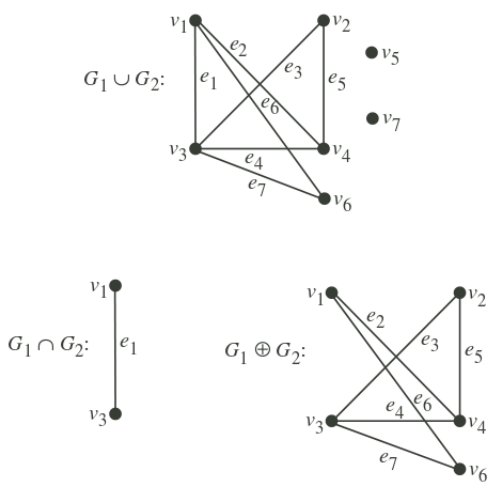


Figure 5.4: Union, intersection and sum, for the above graphs

5.3 Connectivity

While "not connected" is pretty much a dead end, there is much to be said about "how connected" a connected graph is.The simplest approach is to look at how hard it is to disconnect a graph by removing vertices or edges.

Definition 5.4.1 Let $G = (V, E)$ be a graph. Then $\omega(G)$ denotes the number of components of G. If G is a connected graph, then $\omega(G) = 1$. If G is a disconnected graph with k-components, then $\omega(G) = k$.

A vertex v of G is said to be a **cutvertex** or **cutpoint** or **articulation point** if $\omega(Gv) > \omega(G)$. i.e., the removal of v from G *increases* the number of components.

An edge e of G is said to be a **cutedge** or **bridge** if $\omega(G - e) > \omega(G)$, i.e., the removal of e from G *increases* the number of components.

Remark. *If G is connected in which G has a vertex v of degree 1, then the vertex x which is adjacent to v, is cut vertex if and only if $d(x) > 1$.*

Remark. *If G is a graph in which $P : ux_1x_2x_3 \ldots x_nv$ is a unique path connecting u and v, then all internal of $P; x_1, x_2, x_3, \ldots x_n$ are all cut vertices.*

Remark. *Any vertex v lies in a cycle of a graph is not a cut vertex.*

Remark. *Any edge e lies in a cycle of a graph is not a cutedge.*

Definition 5.4.2 Let $G = (V, E)$ be a graph. Let $A \subseteq V$. Then A is said to be a **vertex-cut** if $\omega(GA) > \omega(G)$. i.e., the removal of all vertices. of A from G increases the number of components. Let $B \subseteq E$. Then B is said to be an edge-cut if $\omega(GB) > \omega(G)$. i.e., the removal of all edges of B from G increases the number of components.

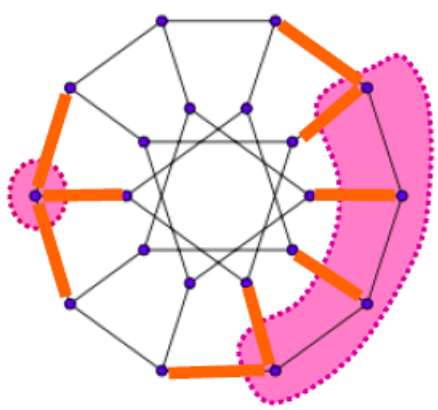


Figure 5.5: The shaded regions are the edge cuts

Remark. *If G is a connected graph in which $d(v) = 2$, $\forall\ v \in V(G)$, then the removal of any two non-adjacent of G disconnects the graph G. i.e., whenever $A = \{u, v\} \subseteq V$ and u is not adjacent to v, then $\omega(G - A) > \omega(G)$.*

Definition 5.4.3 Let $G = (V, E)$ be a graph. Then the **vertex-connectivity** $\kappa(G)$ of G is the minimum k for which G has k-vertex-cut.i.e., $\kappa(G) = min\{k : G$ has k-vertex-cut $\}$.

1. G has k-vertex-cut A with k number of vertices. i.e., $\omega(GA) > \omega(G)$.
2. G has no k'-vertex-cut A' with $k'(< k)$ number of vertices. i.e., $\omega(GA') = \omega(G)$

Similarly, the **edge-connectivity** $\lambda(G)$ of G is the minimum t for which G has t-edge-cut.

In 1932 Hassler Whitney connected all of them into a nice subtle theorem as :-

Theorem 3 (Whitney's)**.** *For any graph G :-*

$$\kappa(G) \leq \lambda(G) \leq \omega(G) \tag{5.1}$$

Some important remarks which can be brought up here are :-

Remark. $\kappa(K_{m,n}) = min\{m, n\}$

Remark. $\kappa(T) = 1$,

Remark. $\kappa(C_n) = 2$,

Remark. $\kappa(P_n) = 1$,

Remark. $\lambda(C_n) = 2$,

Remark. $\kappa(G_1 \cup G_2) = min\{\kappa(G_1), \kappa(G_2)\}$,

5.4 Tree

An **acyclic graph**, one not containing any cycles, is called a **forest**. A connected forest is called a **tree.** (Thus a forest is a graph whose components are trees). The vertex of degree 1 in a tree are it's **leaves**. Every non-trivial tree has a leaf.

Let $G(u, v)$ of a graph on p which may have cycles. A **spanning tree**, T of $G(u, v)$ is a tree, containing all the of G, which mathematically is $V(T) = V(G)$.

Remark. *A tree with at least one edge has at least two pendant edges, or if T is a tree on p $(p \geq 2)$, then T has at least two pendant .*
The Contrapositive. *If G is a graph in which $d(v) \geq 2$, then G contains a cycle.*

A very important theorem for the tree is :-

Theorem 4. *If $T(p, q)$ is a tree with $p \geq 2$ and q edges then,*

$$\boxed{q = p - 1} \tag{5.2}$$

Proof. Using Induction.

The Base case : When $q = 0$, then we have $0 = p - 1 = p = 1$. Thus it's a graph with one vertex. Hence proved.

Inductive Step : Let T be a tree on k number of edges.($q \leq k$) edges. That is, $k = p - 1$ is true.

Induction Hypothesis : To prove if it's true for $q = k + 1$.

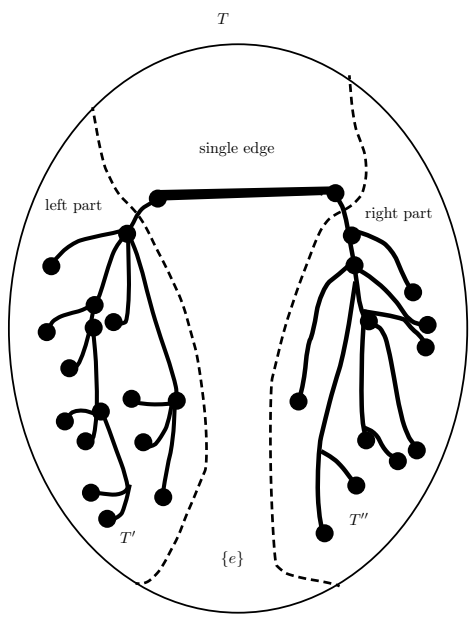


Figure 5.6: $T = T' \cup \{e\} \cup T''$

Using Figure 5.6, we have the left part of the tree as a part of the main tree having k' number of edges, and similarly for the other right part we have a tree having k'' number of edges. Some obvious relation are formed :-

$$p(T) = p(T') + p(T'') \tag{5.3}$$

$$q(T) = q(T') + 1 + q(T'') \tag{5.4}$$

We then apply the assumed correctness to both the parts of the divided tree and get :-

$$q(T') = p(T') - 1 \tag{5.5}$$

$$q(T'') = p(T'') - 1 \tag{5.6}$$

Using equations (5.2) and (5.3) in (5.4) and (5.5), we get :-

$$q(T) - 1 - q(T'') = p(T) - p(T'') - 1 \tag{5.7}$$

$$= q(T) - [p(T'') - 1] = p(T) - p(T'') \tag{5.8}$$

Finally,

$$q(T) = p(T) - 1 \tag{5.9}$$

which proves the induction hypothesis. So, it's proved $\forall\ q \geq 0$. □

Another important theorem can also be proved in a rather elegant way as :-

Theorem 5. *If T is a tree with $p \geq 2$, then T has at least two of degree 1.*

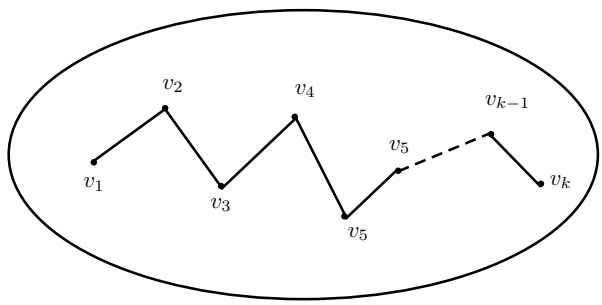


Figure 5.7: Supporting Graph for theorem 5

Proof. Let T be a tree with p $v_1v_2v_3 \ldots v_p$. Then T is a connected graph and T has no cycle. Let $P : v_1v_2v_3 \ldots v_{k-1}v_k$ be the longest path in T. Clearly $d(v_2) \geq 2, d(v_3) \geq 2, \ldots, d(v_{k-1}) \geq 2$. We claim that $d(v_1) = 1$ amd $d(v_k) = 1$. On contrary, suppose $d(v_1) > 1$. The there is another vertex v_s such that v_s is adjacent to v_1 and v_s is none of $v_2v_3v_4, \ldots v_k$ (otherwise , if v_s is anyone of v_i,$(2 \leq i \leq k)$, we get a cycle, a contradiction for T being tree.)

Thus, we have a new path $P' : v_sv_1v_2v_3v_4 \ldots v_{(k-1)}v_k$ is longer path than P, which is a contradiction to our assumption that P is longest path T. Therefore, $d(v_1) = 1$. □

Chapter 6

Planar Graph and the Utility Model

In graph theory, a planar graph is a graph that can be embedded in the plane, i.e., it can be drawn on the plane in such a way that its edges intersect only at their endpoints. In other words, it can be drawn in such a way that no edges cross each other. Such a drawing is called a plane graph or planar embedding of the graph. A plane graph can be defined as a planar graph with a mapping from every node to a point on a plane, and from every edge to a plane curve on that plane, such that the extreme points of each curve are the points mapped from its end nodes, and all curves are disjoint except on their extreme points.

6.1 Euler Formula

An important Formula by Euler in this topic is :-

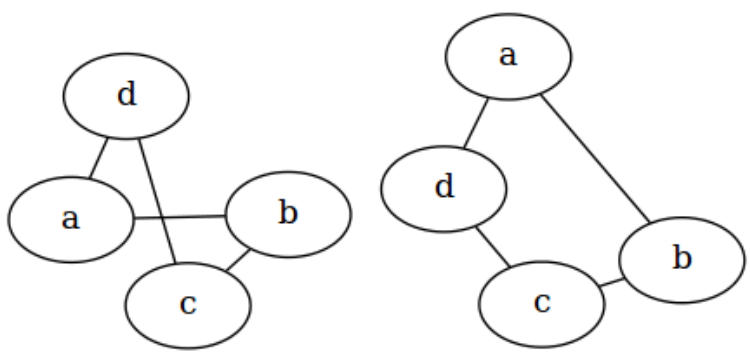


Figure 6.1: Left : Not Planer – Right : Planer

Theorem 6. *Euler's formula* [3] *states that if a finite, connected, planar graph is drawn in the plane without any edge intersections, and v is the number of , e is the number of edges and f is the number of faces (regions bounded by edges, including the outer, infinitely large region), then*

$$v - e + f = 2. \tag{6.1}$$

6.2 The Three Utility Problem

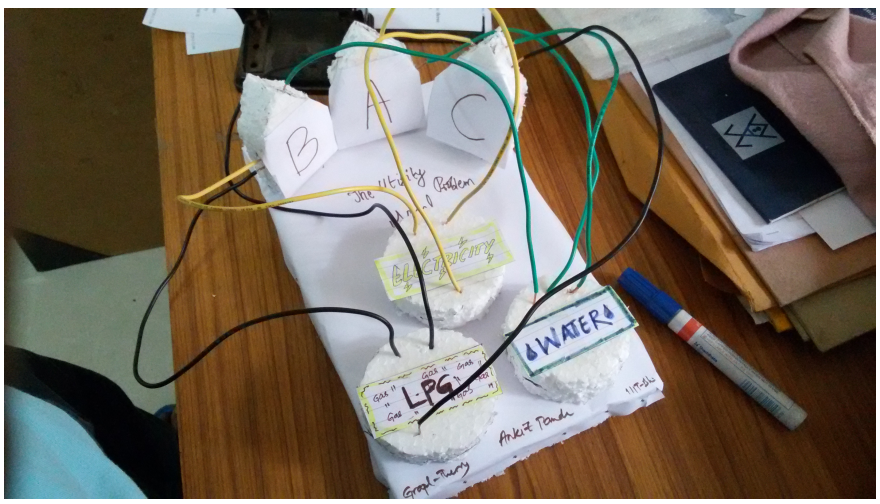


Figure 6.2: The Utility Problem Model

A solution from[3]. It's impossible to connect the houses to the utilities on a flat sheet of paper. In fact, it's impossible in the Euclidean plane (a flat sheet stretched to infinity) and on most other two-dimensional surfaces. Some people have a trick to get around the problem: they send a gas, water, or electric line through one of the houses. Of course, real utility companies don't need tricks. The real world is three-dimensional, so power lines can go over or under each other. One proof of the impossibility of finding a planar embedding of $K_{3,3}$ uses a case analysis involving the Jordan curve theorem, in which one examines different possibilities for the locations of the with respect to the 4-cycles of the graph and shows that they are all inconsistent with a planar embedding.Alternatively, it is possible to show that any bridge less bipartite planar graph with V and E edges has $E \leq 2V4$ by combining the Euler formula $V - E + F = 2$ (where F is the number of faces of a planar embedding) with the observation that the number of faces is at most half the number of edges (because each face has at least four edges and each edge belongs to exactly two faces). In the utility graph, $E = 9$ and $2V4 = 8$, violating this inequality, so the utility graph cannot be planar.

Chapter 7

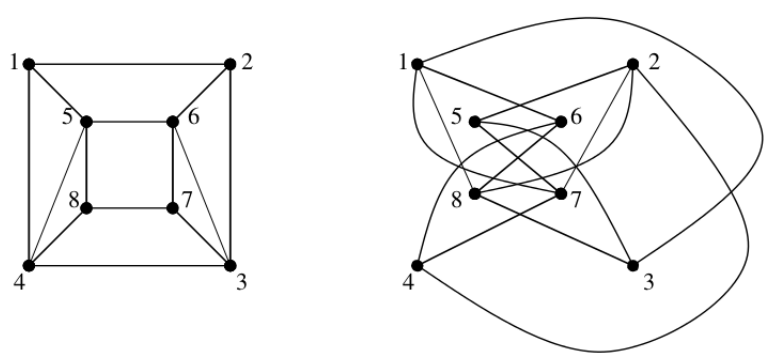
Isomorphism

In graph theory, an isomorphism of graphs G and H is a bijection between the vertex sets of G and H,

$f : V(G) \to V(H)$

such that any two vertices u and v of G are adjacent in G if and only if (u) and (v) are adjacent in H. This kind of bijection is generally called "edge-preserving bijection", in accordance with the general notion of isomorphism being a structure-preserving bijection.

If an isomorphism exists between two graphs, then the graphs are called isomorphic and we write $G \simeq H$. In the case when the bijection is a mapping of a graph onto itself, i.e., when G and H are one and the same graph, the bijection is called an automorphism of G.



Isomorphism: $1 \to 4,\ 2 \to 6,\ 3 \to 1,\ 4 \to 7,\ 5 \to 2,\ 6 \to 8,\ 7 \to 3,\ 8 \to 5.$

Figure 7.1: Isomorphic graph of vertex 8 and it's isomorphisms.

Chapter 8

Conclusion

The summer internship course proved to be indeed helpful for forthcoming works on Discrete Mathematics and Algorithms. Various areas were covered in a short time including Cartesian product of graphs, connectivity, tree,etc. A very short introduction on set was also adequately discussed.

A model for the 3-utilities problem was also made and intuition was developed for the Euler's Formula, which will be helpful in the future for computer science algorithms.

References

[1] Reinhard Diestel, Graph Theory, Springer, 2000

[2] Keijo Rouhonen, Graph Theory, Lecture-Notes, 2013,

[3] Wikipedia, Utility Problem, www.en.wikipedia.org